Flight Management System (FMS) – General Data

1. General

A Flight Management System (FMS) is an integrated system designed to give the pilot centralized control for the navigation sensors, computer based flight planning, fuel and radio management. The FMS functions are provided by the two Advanced Graphics Modules (AGM), located inside the Modular Avionics Unit (MAU) assembly (refer to 31-43-00, Modular Avionics Unit (MAU) (Mod 6/2049) – General Data). The FMS uses two navigation databases, housed in the MAU, containing worldwide or regional data (refer to 45-00-00, Central Maintenance Computer – General Data). A custom database contains flight plans and pilot defined waypoints and an aircraft database contains the aircraft specific configuration for FMS performance calculations.

Using the databases, along with GPS and ADAHRS information, the FMS calculates and displays navigational and flight planning information to the flight crew. The navigation database is commercially produced and updated and released every 28 days.

The FMS provides lateral and vertical control output to the Automatic Flight Control System (AFCS), when S.O.O. 6221 is incorporated.

Provisions have been made by Mod 6/2303 for standard customer option S.O.O. 6313 for a second FMS, providing dual FMS functionality.

Provision have been made by Mod 6/2304 for upgrade of the dual KGS 200 GPS receivers to include SBAS capable units. SBAS Navigation and Approach Capability is provided by standard customer order option S.O.O. 6225.

The FMS interfaces to the following components:

A. MF Controller (KMC 2220) (refer to 31-13-00, Apex Controllers – General Data).

B. Modular Avionics Unit (MAU) (refer to 31-43-00, Modular Avionics Unit (MAU) (Mod 6/2049) – General Data).

C. Air Data, Attitude and Heading Reference System (ADAHRS) (KSG 7200) (refer to 34-23-00, Air Data, Attitude and Heading Reference System (ADAHRS) (KSG 7200) – General Data).

D. Global Positioning System (GPS) (refer to 34-56-00, Global Positioning System (GPS) – General Data).

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